

RASHMI DANGOL

Digital Agronomist | GIS & Remote Sensing Analyst

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CAREER OBJECTIVE

Innovative Digital Agronomist and FAA-certified Remote Pilot with a major in agriscience and a Graduate Certificate in GIS. Seeking extensive experience in UAV-based remote sensing and geospatial analysis and modeling to drive data-driven decisions in industry. Proven track record at **Syngenta** and **Bayer Crop Science** in optimizing product placement, high-throughput phenotyping, and automated field mapping to enhance crop performance and yield outcomes.

TECHNICAL SKILLS

- **Geospatial & Remote Sensing:** ArcGIS Pro, ArcGIS Online, ENVI, Pix4D Mapper, ArcGIS Experience Builder, Satellite Imagery Analysis, Photogrammetry.
- **Programming & Data Science:** Python (ArcPy, GeoPandas), Computer Vision (YOLO), Deep Learning, SAS, SPSS, Microsoft Power BI.
- **Field Operations:** FAA Part 107 Certified Pilot, DJI Matrice 300 RTK, GPS Rover, High-Throughput Phenotyping, Soil and Tissue Sampling.
- **Industry Tools:** Smart Farm Plus, Self-Service Shape, Digital Field Mapping, Yield Data Analytics.

PROFESSIONAL EXPERIENCE

Product Placement Phenotyping Intern | Syngenta, Malta, IL | May 2025 – Aug 2025

- **Geospatial Mapping:** Engineered digital field maps using Google Earth Pro and Self-Service Shape for precise project planning and spatial analysis of r.
- **UAV Operations:** Led flight operations using DJI Matrice 300 RTK drone to capture high resolution imagery for early-season vigor and corn emergence quality.
- **Data Pipeline Development:** Collaborated on a Python-based machine-learning image processing pipeline to generate first-model estimators for crop health indices.
- **Insight Generation:** Delivered actionable geospatial insights to the product placement team to optimize variety selection based on spatial variability.

Research Assistant (Precision Agriculture) | Texas A&M University | Illinois State University | May 2022 – Apr 2026

- **Remote Sensing Research:** Conducted large-scale experiments utilizing satellite and UAV imagery to evaluate pasture management.
- **AI Integration:** Developed and trained **YOLO computer vision models** for automated weed detection, reducing manual scouting time and improving precision in management.
- **Workflow Automation:** Created custom Python functions using **ArcPy** to automate geospatial data processing, increasing efficiency in time-series analysis.
- **Flood Analysis & Geospatial Analysis:** Conducted flood analysis for Kerr County using soil volumetric content and land cover data in **ArcGIS Pro** and **ArcGIS Notebooks**.
- **Advanced Analytics:** Performed image mosaicking, calibration, and vegetative index calculations (NDVI, NDWI) to monitor forest health.

Seed Technology Intern | Bayer Crop Science, Oxnard, CA | May 2023 – Aug 2023

- **Yield Analysis:** Executed field data collection for yield performance using GPS rovers, ensuring high data integrity for the LATAM seeds technical sheets.
- **Data Coordination:** Managed soil analysis and tissue culture data via the Smart Farm Plus application to support seed quality protocols.

Digital Enumerator/Surveyor | CIMMYT, Nepal | Jun 2018 – Aug 2018

- **Survey Leadership and Data Collection:** Led a 6-member team for the National Seed and Fertilizer Project (NSAF), leveraging KoboToolbox to digitize data collection and evaluate the effectiveness of ICT tools among rural smallholder farmers.

EDUCATION

- **Graduate Certificate in GIS** | Texas A&M University, College Station, TX | *Dec 2025*
- **Master of Science in Agriscience** | Illinois State University, Normal, IL | *May 2024*
- **Bachelor of Science in Agriculture** | Agriculture and Forestry University, Nepal | *Sep 2019*

CERTIFICATIONS & AWARDS

- **FAA Part 107 Certified Remote Pilot** (Commercial Drone License) | *Sep 2024*
- **Visualizing Data with Python** | IBM | *Jun 2025*
- **James L. Fisher Outstanding Thesis Award** | Illinois State University | *Nov 2024*
- **Outstanding Graduate Student Recognition** | Illinois State University | *Apr 2024*